

Code-Layout, Readability and Reusability

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Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

Well-structured code is essential for maintainability and collaboration. This section documents the standards, patterns, and principles followed to ensure the codebase (data cleaning scripts and the Flask web application) is clean, readable, and highly reusable.

Code Quality & Structure Details

S.No	Quality Principle	Implementation Details / Description	Specific Project Examples (Files/Folders)
1	Folder/Directory Structure	The Flask app separates templates (one HTML page per route) from static assets (exported chart images), while the data-cleaning work lives in its own script separate from the Tableau workbook.	/templates (home.html, about.html, dashboard.html, storyboard.html, base.html), /static/images
2	Naming Conventions	snake_case used for Python variables and functions in the cleaning script; lowercase-with-underscores for exported chart image filenames; template filenames match their route/page name.	clean_text(), countries_per_region.png, danger_sites_analysis.png, dashboard.html
3	Code Readability & Comments	The cleaning script documents each transformation step inline (HTML unescape, tag stripping, whitespace normalization) so the logic is auditable without re-deriving it from the output.	clean_text() function in the data-preprocessing script
4	Modularity & Reusability	Cleaning logic is isolated into a single reusable clean_text() function applied identically across Site Name, Description, and Justification columns; Flask pages extend one shared base layout instead of duplicating markup.	base.html extended by home.html, about.html, dashboard.html, storyboard.html
5	Formatting & Linting Tools	Consistent PEP 8-style formatting in Python scripts; dependencies pinned via a requirements file for a reproducible environment.	requirements.txt
6	Error Handling Patterns	File I/O and data-parsing steps in	Data validation checks in the

		the cleaning script are wrapped in try/except blocks; the cleaned workbook is verified (zero formula errors, zero duplicates) before being connected to Tableau.	preprocessing script; app.py route handling
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